Report Biomimicry Challenge

ZHAW Institute for Sustainable Development | Blockweek Bionics 2026

Challenge 1 — Urban Heat Islands

Cooling the City

"Picture"

Insert a concept image here:
\includegraphics[width=\linewidth]{figures/cover.jpg}

Students	Contribution to the report / (sections)
[TODO: Name 1]	[TODO: e.g. Sections 2, 3.2, engineering calculations]
[TODO: Name 2]	[TODO: e.g. Sections 3.1, 4, biological research]
[TODO: Name 3]	[TODO: e.g. Sections 1, 5, figures and editing]
[TODO: Name 4]	[TODO: e.g. Section 3.3, measurements, appendix]

Comments	[TODO: optional remarks for the instructors]

Supervisors: Jörg Musiolik, Corinna Baumgartner | Submission date: [TODO: date]

Contents

1 Introduction

1.1 Background and relevance

[TODO: 2–3 paragraphs. Suggested content, to be written out and referenced:]

- Urban heat islands (UHI): built-up areas are warmer than their rural surroundings, mainly because impervious surfaces absorb short-wave radiation, store the heat and release it at night, while evapotranspiration and long-wave loss to the sky are reduced (Oke 1982).
- Relevance for Switzerland: number of hot days and tropical nights, health impact on residents of upper floors without air conditioning, and the fact that mechanical cooling increases both electricity demand and waste heat in the street canyon. **[TODO: add local figures and cite]**
- Policy context: the City of Zurich has an explicit heat-mitigation plan (Stadt Zürich 2020), which makes retrofit measures on existing building envelopes a realistic field of application.

1.2 Aim of the report and intended contribution

[TODO: State the aim in 3–5 sentences, e.g.:] This report documents a biomimicry design process for reducing heat load on the envelope of existing urban buildings. It follows the Biomimicry Design Spiral (Baumeister et al. 2014) through scoping, biological discovery and abstraction, engineering concept development, validation and evaluation. The intended contribution is a technically specified, quantitatively checked envelope concept, together with a transparent account of which biological mechanisms were transferred, which were rejected, and on what evidence.

1.3 Structure of the report

[TODO: One short paragraph describing Sections 2–5.]

2 Challenge & Scoping

2.1 Problem definition and design question

Given challenge (Challenge 1). *Cities are facing increasing heat stress due to climate change. Develop bioinspired strategies to reduce urban heat through building materials, façade design, or urban planning.*

The challenge as given spans three different scales — material, building component and city planning. Treating all three would make the project broader rather than more focused, which the systems view in Section 2.3 is explicitly meant to prevent. We therefore narrow the challenge to the scale at which our team can actually design, calculate and test something within the available time: the *outer surface of an existing building envelope*.

A second reason for this cut is physical. The urban heat island is driven by the surface energy balance: solar radiation absorbed by opaque urban surfaces during the day is stored and re-emitted as sensible heat, while long-wave loss to the sky is reduced by the limited sky view factor of the street canyon (Oke 1982). The building envelope is one of the largest of these surfaces and, unlike streets or trees, it is directly accessible to a design intervention by a single building owner.

Design question (solution-neutral):

How might we keep the outer envelope of existing urban buildings — and the air and surfaces next to it — cool during summer heat waves, using only passive means and locally available water?

The question was deliberately formulated *after* the context, systems and function analysis, not before. It names no solution (no coating, no greening, no ventilation principle), states the operating context (existing buildings, summer heat waves), and fixes the two hard constraints that follow from Section 2.2 (no auxiliary energy, no potable water). Following the guidance on scoping questions, it sits between "How might we cool the city?" (too broad — no defined intervention point) and "How might we design a reflective roof paint?" (too narrow — presupposes the solution).

2.2 Context and constraints

Where and for whom. A Swiss midland city, taken as a reference case (Zurich). Dense perimeter-block development with narrow street canyons and a high share of sealed surfaces. The system users are the residents of the upper floors, who are exposed to the strongest heat load and usually have no mechanical cooling; the pedestrians in the street canyon, who are exposed to radiation from the façade; and the building owner, who decides on and pays for the retrofit.

When. The design must perform in two distinct operating windows, which the systems view made explicit:

- **Day window (approx. 11:00–18:00, clear sky):** peak solar irradiance on the order of 950 W/m^2 on a horizontal surface and air temperatures up to about 36 °C. Dominant problem: absorbed short-wave radiation. **[TODO: replace with measured values from MeteoSwiss for your reference site]**
- **Night window (approx. 22:00–06:00):** the stored heat is released again, producing tropical nights. Dominant problem: *delayed* release of stored heat and a reduced sky view factor that limits radiative loss. A concept that only performs during the day does not solve the urban heat island.

Constraints. Table 1 lists the constraints the concept has to satisfy. They are separated into hard constraints (a violation disqualifies a concept) and soft constraints (they influence the evaluation in Section 3.3).

2.3 System view and project boundary

Figure 1 places the challenge in its system hierarchy. The purpose of the hierarchy is to prevent tunnel vision; the purpose of the boundary drawn afterwards is to prevent scope explosion.

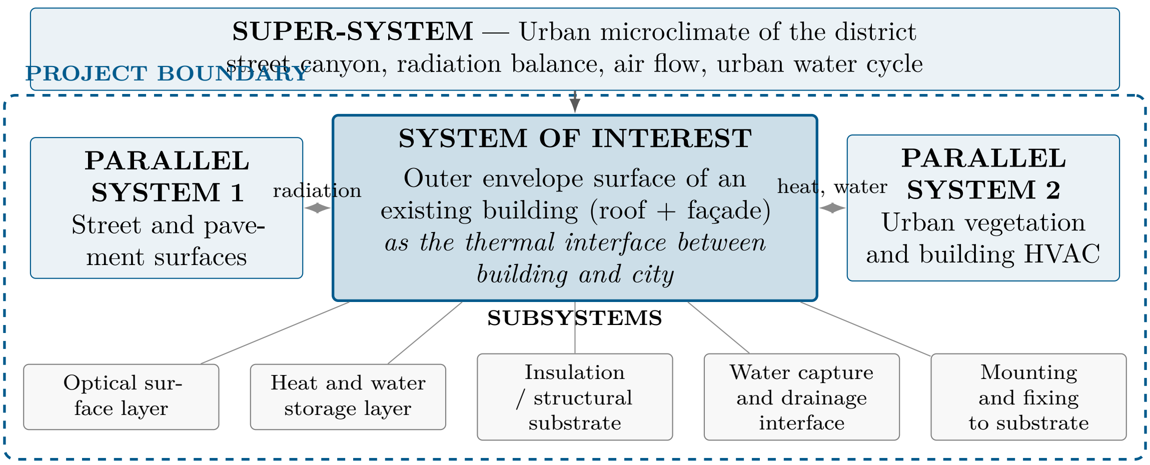

Figure 1: System view for Challenge 1. The dashed line is the project boundary: everything inside it is designed and analysed by the team.

Table 1: Context conditions and constraints.

Type	Class	Constraint
Energy	hard	No auxiliary electrical energy for the cooling function itself (pumps, fans, chillers). Sensors for validation are excluded from this constraint.
Water	hard	No potable water. Only rainwater harvested on site or atmospheric moisture may be used.
Existing building	hard	Retrofit onto an existing envelope. The load-bearing structure, floor plan and window openings are not redesigned.
Climate	hard	Must survive the Swiss annual cycle: frost down to about $-10\,°C$, freeze–thaw cycles, driving rain, UV exposure. A summer-only solution is not acceptable.
Winter	hard	Must not increase the heating demand appreciably (a high solar reflectance also reduces useful winter solar gains — this trade-off must be quantified, not ignored).
Regulatory	soft	Fire regulations for façade build-ups; heritage protection may restrict visible changes on street-facing façades.
Structural	soft	Limited additional dead load and build-up thickness on an existing façade.
Maintenance	soft	Soiling, algae growth and weathering degrade the optical function; the concept should keep working with little maintenance over $\geq$ 20 years.
Economic	soft	Cost per square metre in the range of a normal façade or roof refurbishment. **[TODO: find a benchmark figure and cite it]**

Table 2: Project boundary: what is deliberately in and out of the project.

IN the project	OUT of the project
Passive control of the short-wave and long-wave radiation balance of the envelope surface	Redesign of the building itself: structure, floor plan, windows, insulation thickness
Heat and water storage behaviour of the outermost build-up, including the night-time release	Active building services: chillers, mechanical ventilation, heat pumps, indoor comfort control
Capture and controlled release of rainwater within the envelope build-up	Street trees, parks, urban greening at district level, urban planning
Interface with the street canyon (radiation towards pedestrians) and with the night sky	Municipal water supply and treatment, sewage system
Mounting concept on an existing substrate and durability of the optical function	Permitting, heritage approval, business model, full life cycle assessment

What the systems view added. *The systems view changed our project scope because* it showed that the envelope does not only exchange heat with the building interior, but also with two systems we had not considered: the pedestrian level of the street canyon (the façade radiates onto people) and the night sky (the only heat sink that is available without energy input). This added one function (*emit long-wave radiation to the sky*), one constraint (the concept must also perform at night, not just at the daytime peak) and one KPI (net radiative cooling power at night, KPI 5 in Table 4). The comparison with the parallel system "urban vegetation" also sharpened a constraint: vegetation achieves its cooling through evapotranspiration and therefore needs a continuous water supply, which in a heat wave is exactly the resource that is scarce. This is the origin of the hard constraint "no potable water".

2.4 Functions and functional diagram

Following the scoping method, the functions are formulated as solution-neutral verbs and answer the question "what must the design *do*?", not "what do we want to build?". The verbs are aligned with the Biomimicry Taxonomy so that they can be used directly as search terms in AskNature (Biomimicry Institute n.d.).

Table 3: Function list. Main function, sub-functions and supporting functions.

ID	Function (verb)	Explanation / taxonomy link
Main function		
F0	Regulate heat gain of a surface	Modify / regulate physical state: manage temperature
Sub-functions — energy path		
F1	Reflect short-wave radiation	Prevent absorption of solar energy in the first place
F2	Emit long-wave radiation	Release heat to the sky, in particular in the atmospheric window
F3	Transfer heat to ambient air	Convective dissipation at the surface
F4	Store thermal energy temporarily	Buffer and shift the daily peak
F5	Shade a surface	Reduce the irradiated area, e.g. through surface geometry
Sub-functions — water path		
F6	Capture water	Collect rain or atmospheric moisture
F7	Retain water	Store it until it is needed during the heat peak
F8	Release water in a controlled way	Meter evaporation instead of losing the reserve at once
Supporting functions		
F9	Keep a surface clean	Maintain the optical function against soiling and algae
F10	Resist degradation / self-repair	Withstand UV, frost and weathering over the service life
F11	Attach to a substrate	Fix the build-up to an existing wall or roof

Figure 2 shows the functional diagram. It is kept solution-neutral: it contains only functions and the energy and material flows between them, no components and no material choices.

2.5 Biologized questions

The functions are translated into questions to nature, which form the search strategy for Section 3.1:

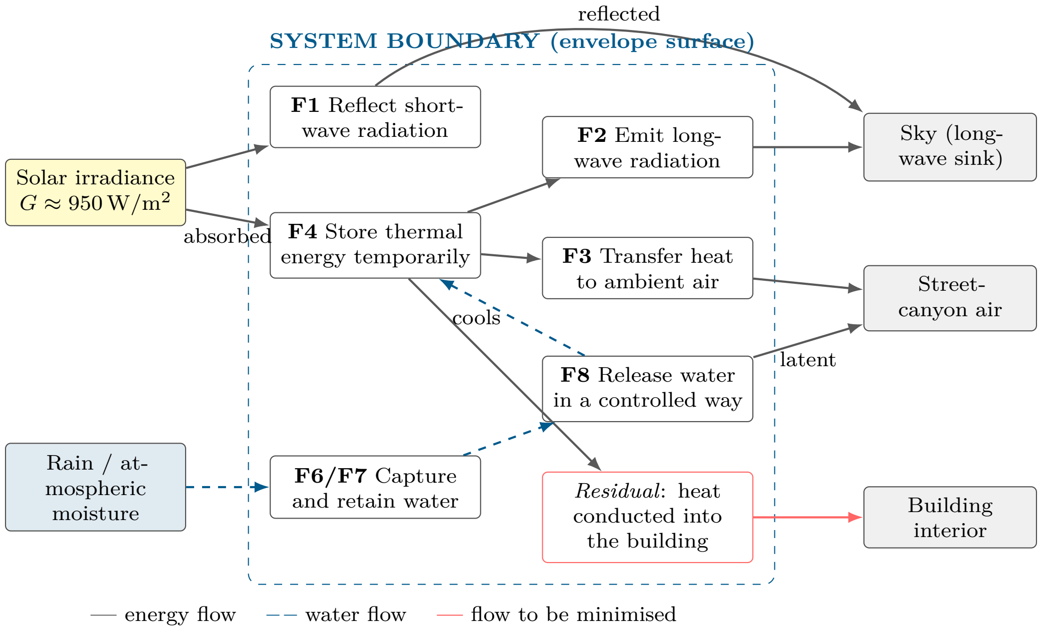

Figure 2: Solution-neutral functional diagram of the envelope surface. Supporting functions F9–F11 (keep clean, resist degradation, attach) act on all functions inside the boundary and are omitted here for readability.

- How does nature reflect intense solar radiation without pigments that bleach — and how does it stay white or bright over a lifetime?
- How does nature release excess heat to the night sky?
- How does nature dissipate heat where water is scarce?
- How does nature capture, store and meter out small amounts of water?
- How does nature buffer a daily temperature extreme instead of resisting it?
- How does nature keep a functional surface clean without cleaning it?
- How does nature repair a surface that is damaged by weathering?

2.6 Performance criteria and KPIs

The KPIs are derived from the functions, interfaces and constraints identified above — not invented independently of them. They are used twice: to guide the engineering checks in Section 3.2 and to compare the concepts in Section 3.3.

Note on the targets. The target values above are the team's *aspirational* benchmarks set during scoping. They are hypotheses, not results. Every one of them is checked in Section 3.2 and either confirmed, revised or explicitly declared unverified. **[TODO: before hand-in: verify each cited benchmark value against the primary source — the rubric penalises unchecked claims]**

Important distinction. KPI 3 measures a *surface* temperature, not the air temperature of the city. A high-reflectance surface can easily be 20 K cooler than a dark one, while the effect on the air temperature of the district is smaller by roughly two orders of magnitude (Santamouris 2014; Akbari et al. 2009). Conflating the two is the most common error in urban-heat-island projects and would invalidate our evaluation, so both quantities are reported separately throughout this report.

Table 4: Performance criteria. Function → KPI → unit → target. Reference case: an existing flat roof / plastered façade with a solar reflectance of about 0.15–0.25.

#	Function	KPI	Unit	Target / benchmark
1	F1 Reflect short-wave radiation	Solar reflectance α_{sol} (300–2500 nm)	–	$\geq$ 0.80. Cool-roof products typically 0.65–0.85; ultra-white laboratory paints reach $\approx$ 0.98 (Li et al. 2021).
2	F2 Emit long-wave radiation	Thermal emittance ε in the atmospheric window (8–13 µm)	–	$\geq$ 0.90. Required for sub-ambient radiative cooling (Raman et al. 2014).
3	F0 Regulate heat gain	Peak surface temperature reduction ΔT_s vs. reference, clear sky, 14:00	K	$\geq$ 15 K. **[TODO: confirm against your own measurement or simulation — do not claim this before you have checked it]**
4	F0 / F4	Heat conducted into the building per hot day	$\mathrm{W\,h/(m^2\,d)}$	$\geq$ 40 % below the reference case.
5	F2 (night)	Net radiative cooling power at 02:00, clear sky	$\mathrm{W/m^2}$	$\geq 40\,\mathrm{W/m^2}$; reported values for radiative coolers are of this order (Raman et al. 2014; Zhai et al. 2017).
6	F6–F8 Water path	Potable water consumption	$\mathrm{L/(m^2\,d)}$	= 0 (hard constraint). Harvested rainwater use to be reported separately.
7	F9/F10 Durability	Retention of α_{sol} after soiling / weathering	% of initial	$\geq$ 90 % after three years. Aged cool roofs commonly lose a significant fraction of their initial reflectance (Santamouris 2014).

2.7 Relevant Life's Principles

The Life's Principles that are most relevant to this specific challenge, and that are used again as a sustainability benchmark in Section 4:

- **Use low energy processes** — the entire cooling function must be passive; this is the hard energy constraint.
- **Be locally attuned and responsive** — the concept must respond to the local diurnal cycle and the local water availability, not to an average annual climate.
- **Be resource efficient (material and energy)** — fit form to function, low added mass per square metre (see also KPI 7).
- **Adapt to changing conditions** — performance is needed in a summer heat wave, but the same surface must not harm the winter energy balance.
- **Maintain integrity through self-renewal** — the optical function degrades through soiling; a self-cleaning or self-renewing surface preserves the KPI over the service life.
- **Use life-friendly chemistry** — relevant when selecting pigments, binders and biocides for the outer layer.

Quality check (slide 29).

- ☐ Is our system of interest clearly defined? — Yes: the outer envelope surface of an existing building.
- ☐ Is our project boundary manageable? — Yes, see Table 2.
- ☐ Are our functions solution-neutral? — Yes, verbs only; no components named in Table 3 or Figure 2.
- ☐ Did the systems view reveal additional functions or constraints? — Yes: F2 (emit to the sky), the night operating window, and the water-scarcity constraint.

- ☐ Are our most important functions measurable? — Yes, KPIs 1–7.
- ☐ Is the "How might we. . . ?" question open to different solutions? — Yes: radiative, evaporative and geometric strategies all remain possible.

3 Bio-Inspired Design Process

[TODO: One short paragraph: how the team organised the search and the design path, so the reasoning chain from Section 2 to Section 3.3 is visible.]

3.1 Discovering & Abstracting

3.1.1 Research path

[TODO: Describe how you searched: AskNature functions used, search terms, databases, which biologized question from Section 2.5 led to which model.]

Table 5: Biological models found, and the selection / de-selection decision.

Organism / model	Function	Mechanism (short)	Selected?
Saharan silver ant *Cataglyphis bombycina*	F1, F2	Triangular hairs scatter visible and near-infrared light and simultaneously enhance emissivity in the mid-infrared (Shi et al. 2015)	**[TODO: yes/no + why]**
White beetle *Cyphochilus*	F1	Brilliant whiteness from a disordered chitin network only a few micrometres thick, without pigment (Vukusic et al. 2007)	**[TODO: yes/no + why]**
African elephant skin	F6–F8	Network of micrometre-scale cracks retains water and extends evaporative cooling (Martins et al. 2018)	**[TODO: yes/no + why]**
Desert shrub *Encelia farinosa*	F1, F5	Dense leaf pubescence raises reflectance and is adjusted seasonally (Ehleringer and Mooney 1978)	**[TODO: yes/no + why]**
Termite mound	F3	Classic biomimicry reference, but the "chimney air-conditioning" account is contested; measurements support wind-driven transient gas exchange rather than a thermosiphon (Turner and Soar 2008)	**[TODO: recommended: DE-SELECT with this justification]**
[TODO: further models]			

3.1.2 Mechanisms in depth

[TODO: For the 2–3 selected models: explain the mechanism properly — geometry, length scales, physics, and the evidence. Add a sketch. This is worth 20 % of the grade and descriptive text alone is not enough.]

3.1.3 Abstraction into design principles

[TODO: Fill in. Keep the three columns strictly separate.]

Table 6: From biological observation to transferable design principle.

Biological observation	**Abstracted principle**	**Possible technical interpretation**
[TODO: what was observed]	[TODO: the principle, free of the organism]	[TODO: how it could be built — not yet a decision]

3.2 Creating: Engineering Concept Development

3.2.1 Ideation

[TODO: Document the method (Crazy 8s / morphological matrix). Include a photo or scan of the sheets. Show at least three initial combinations.]

3.2.2 Concept A — [TODO: name]

[TODO: Annotated drawing + description: components, functional chain, flows, candidate materials, key technical parameters.]

3.2.3 Concept B — [TODO: name]

[TODO: Must be technically DISTINCT from A — a different physical principle, not a variant. E.g. a radiative (dry) concept vs. an evaporative or geometric (self-shading) concept.]

3.2.4 Engineering Canvas and critical questions

[TODO: Insert the Engineering Canvas. Then derive 2–3 critical engineering questions of the form: "What must be true for the concept to work?"]

1. **[TODO: Critical question 1]**
2. **[TODO: Critical question 2]**
3. **[TODO: Critical question 3]**

3.2.5 Engineering check 1: steady-state surface energy balance

[TODO: This check is prepared because it is the most direct way to test KPI 1 and KPI 3. Fill in your own numbers, state every assumption, and keep the units.]

For an opaque surface in steady state, the energy balance per unit area is

$$(1 - \alpha_{\mathrm{sol}})\, G \;+\; \varepsilon L_{\downarrow} \;-\; \varepsilon \sigma T_s^4 \;-\; h_c\,(T_s - T_a) \;-\; q_{\mathrm{cond}} \;=\; 0, \tag{1}$$

where α_{sol} is the solar reflectance, G the global irradiance on the surface, ε the thermal emittance, $L_{\downarrow}$ the incoming long-wave radiation from the sky, $\sigma = 5.67 \times 10^{-8}\,\mathrm{W/(m^2\,K^4)}$ the Stefan–Boltzmann constant, T_s the surface temperature, T_a the air temperature, h_c the convective heat transfer coefficient and q_{cond} the heat conducted into the construction.

Result and meaning. **[TODO: Solve Eq. (1) for T_s for the reference and for the concept, report both, and state explicitly what the difference does and does *not* prove (see the note on KPI 3 in Section 2.6).]**

Table 7: Assumptions for the energy balance. Every value needs a source.

Quantity	**Value used**	**Justification / source**
G	[TODO:] W/m^2	[TODO: MeteoSwiss, clear-sky summer day]
T_a	[TODO:] °C	[TODO:]
h_c	[TODO:] W/(m^2 K)	[TODO: low wind speed in a street canyon]
$L_\downarrow$	[TODO:] W/m^2	[TODO: clear-sky estimate; state the correlation used]
α_{sol} reference	[TODO:]	[TODO: aged bitumen / plaster]
α_{sol} concept	[TODO:]	[TODO:]

3.2.6 Engineering check 2: [TODO: title]

[TODO: Options that are realistic within the week: (a) a simple outdoor measurement — identical plates with different surfaces, logged with a thermometer or a temperature sensor over a few hours; (b) a transient 1-D model of the build-up over a diurnal cycle; (c) a water balance — litres of rainwater per square metre per hot day against the latent heat of vaporisation of 2.45 MJ/kg; (d) a material and cost comparison. Report method, raw data, result and uncertainty.]

3.3 Concept Evaluation & Iteration

Table 8: Decision matrix. Ratings 1–5, weights from Section 2.6. Every rating must be justified in the text.

KPI	**Weight**	**Concept A**	**Concept B**	**Evidence used**
1 – Solar reflectance	[TODO:]	[TODO:]	[TODO:]	[TODO:]
2 – Thermal emittance	[TODO:]	[TODO:]	[TODO:]	[TODO:]
3 – Peak surface temperature	[TODO:]	[TODO:]	[TODO:]	[TODO:]
4 – Heat into the building	[TODO:]	[TODO:]	[TODO:]	[TODO:]
5 – Night radiative cooling	[TODO:]	[TODO:]	[TODO:]	[TODO:]
6 – Potable water use	[TODO:]	[TODO:]	[TODO:]	[TODO:]
7 – Durability of the optics	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Weighted total		[TODO:]	[TODO:]	

3.3.1 Review and feedback

[TODO: World café peer feedback, instructor feedback, and AI used as a critical reviewer — including where the team disagreed with the AI and why. This belongs in Appendix A as well.]

3.3.2 Selection and Final Concept V2

[TODO: State which concept (or justified hybrid) was selected and describe V2 in detail.]

Table 9: Final output of the creating and evaluating phase: Concept V1a/b → engineering review → evidence → Concept V2.

What changed	Why it changed	Evidence that informed the change
[TODO:]	[TODO:]	[TODO:]
[TODO:]	[TODO:]	[TODO:]

3.3.3 Remaining uncertainties

[TODO: What is still unproven, and what should be tested or developed next.]

4 Sustainability & Life's Principles

4.1 Application of Life's Principles to the final concept

[TODO: Take the principles from Section 2.7 and assess each one honestly: how well does the final concept apply it, and where does it fall short? A table with a short assessment per principle works well. Listing principles without assessing them is explicitly marked down in the rubric.]

4.2 Environmental benefits, limitations and trade-offs

[TODO: Quantify where possible. The trade-offs that are already visible for this challenge:]

- **Summer vs. winter:** a high solar reflectance reduces summer heat gain but also reduces useful passive solar gains in winter. **[TODO: estimate both and give the net balance for the Swiss climate]**
- **Cooling vs. water:** evaporative cooling costs roughly 2.45 MJ per kilogram of water evaporated. **[TODO: convert this into litres per square metre per hot day and check it against the rainwater actually available on site — during a heat wave it will not rain, which is precisely the point of the storage function F7]**
- **Local vs. global:** the concept moves heat out of the street canyon; state clearly whether it reduces total energy demand or only redistributes heat.
- **Materials:** **[TODO: pigments, binders, biocides against algae, end of life, separability of the layers, recyclability]**

4.3 Influence of sustainability considerations on the design

[TODO: Name at least one concrete design decision that was changed because of a sustainability consideration — otherwise this section reads as an add-on.]

5 Discussion & Conclusion

5.1 Contribution of the biomimicry process

[TODO: Be concrete and honest. Which idea came specifically from the biological model, and which would a conventional engineering approach have produced anyway? A well-argued "this part was conventional" is worth more than an overstated claim.]

5.2 Strengths and weaknesses of the final concept

[TODO:]

5.3 Assessment against the challenge and the KPIs

[TODO: Go back through KPIs 1–7 from Table 4 and state for each one whether it was met, missed, or could not be assessed with the available evidence.]

5.4 Innovative character and limits of the evidence

[TODO: State clearly what has been demonstrated (calculation, measurement, literature) and what remains an assumption.]

6 References

Akbari, Hashem, Surabi Menon, and Arthur Rosenfeld (2009). "Global cooling: Increasing world-wide urban albedos to offset CO2". In: *Climatic Change* 94.3, pp. 275–286. DOI: 10.1007/s10584-008-9515-9.

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Santamouris, M. (2014). "Cooling the cities — A review of reflective and green roof mitigation technologies to fight heat island and improve comfort in urban environments". In: *Solar Energy* 103, pp. 682–703. DOI: 10.1016/j.solener.2012.07.003.

Shi, Norman Nan et al. (2015). "Keeping cool: Enhanced optical reflection and radiative heat dissipation in Saharan silver ants". In: *Science* 349.6245, pp. 298–301. DOI: 10.1126/science.aab3564.

Stadt Zürich (2020). *Fachplanung Hitzeminderung*. Tiefbau- und Entsorgungsdepartement, Stadt Zürich. **[TODO: verify title, year and exact publisher before hand-in]**.

Turner, J. Scott and Rupert C. Soar (2008). "Beyond biomimicry: What termites can tell us about realizing the living building". In: *Proceedings of the First International Conference on Industrialized, Intelligent Construction (I3CON)*. Critical re-assessment of the termite-mound ventilation narrative — read before citing the Eastgate Centre example. Loughborough University.

Vukusic, Pete, Benny Hallam, and Joe Noyes (2007). "Brilliant whiteness in ultrathin beetle scales". In: *Science* 315.5810, p. 348. DOI: 10.1126/science.1134666.

Zhai, Yao et al. (2017). "Scalable-manufactured randomized glass-polymer hybrid metamaterial for daytime radiative cooling". In: *Science* 355.6329, pp. 1062–1066. DOI: 10.1126/science.aai7899.

Appendix A – AI Use Statement

Relevant AI-supported activities are documented below. Not every prompt is listed; the focus is on how AI contributed to the work and how important outputs were verified. AI output is not used as evidence for factual claims.

Table A.1: AI use statement.

AI tool	**Used for**	**How the output influenced the project**	**How important output was verified**
AskNature Chat	Discovering biological strategies for functions F1–F8	**[TODO: e.g. generated candidate organisms for further research]**	**[TODO: AskNature profile + peer-reviewed paper]**
[TODO: LLM assistant]	**[TODO: e.g. structuring the scoping, drafting the LaTeX report template, critical review of the concepts]**	**[TODO: e.g. produced the first draft of the function list, which the team revised]**	**[TODO: e.g. every biological and physical claim checked against the cited primary sources; benchmark values recomputed by hand]**

Appendix B – Optional Supporting Material

[TODO: Additional calculations, extended concept drawings, raw measurement data or prompt excerpts where useful.]